

NinaXander: Composing Frozen Language Models Across Model Families via a Shared Latent Representation — Conditions and Limits

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Abstract—NinaXander is a language model that connects frozen layers from different pretraining families through a shared-latent adapter. Using RWKV-4-Raven-7B (recurrent) and Tulu-Pythia-6.9b (Transformer), we test whether representations at corresponding layers can be connected by a simple transformation, and whether that connection supports functional composition. There are four main results. (i) Under matched conditions — a shared tokenizer, width, depth, and training domain — the centered latent correlation reaches $\rho_{\text{ctr}}=0.901$. Shuffling the row correspondence destroys the alignment, yet the CKA of the raw representations does not peak at same-numbered layers, and performance degrades sharply under a domain shift to WikiText. The result is therefore not a general semantic space but a token-aligned representation that is learnable under favorable conditions. (ii) The four-path read-outs are 0.695/0.559/0.711/0.590 for RWKV self-recon/RWKV→Pythia/Pythia self-recon/Pythia→RWKV, so reconstruction error remains even with compression removed. (iii) The best direct affine maps reach 0.487/0.535 for RWKV→Pythia/Pythia→RWKV, while 13.4%/12.0% of the trained adapter’s output cannot be explained by any affine map; disabling the same adapter’s nonlinear branch collapses both cross read-outs to negative R^2 . (iv) Chimeras composed at a single boundary produce syntactically plausible generations and perform multiple choice, but do not reach the stronger parent. Pythia→RWKV exceeds RWKV→Pythia at deep boundaries by 8.9–18.8 accuracy points at the same layer, and the Pythia-prefix→RWKV-suffix Pythia→RWKV ($L=4$) keeps only 5 Transformer layers, cutting KV by 84.4% while retaining ARC-Easy 62.2% and SciQ 69.2%. It does not, however, significantly exceed RWKV, and the released default is itself a post-hoc selection on the same two test sets. Removing unused blocks reduces resident weights to 14.11 GiB, but out-of-domain WikiText and the speed of the reference implementation remain substantial problems.

I. Introduction

This work constructs NinaXander¹—a chimera model that connects frozen blocks from different families through

¹NinaXander derives from the chimera in Fullmetal Alchemist—a being formed by fusing the girl Nina and the dog Alexander. Even though the synthesis succeeds and the fusion functions, it does not match the original beings. The chimera in this work likewise operates but does not reach the stronger parent, and the name alludes to this correspondence.

a single trained adapter—and clarifies the conditions under which it succeeds and its limits. Recombining heterogeneous blocks requires a mechanism that hands residual representations across families while preserving model function.

To this end, this work introduces a shared-latent adapter and examines two questions. First, does a shared representation exist between corresponding layers of heterogeneous models that can be connected by a low-depth mapping? Second, how much function does a chimera built on that representation retain relative to the parent models, and how much inference-memory reduction does it achieve? The contributions of this work are (i) alignment measurement with the common mode removed, (ii) a controlled experiment comparing the four paths RWKV self-recon/RWKV→Pythia/Pythia self-recon/Pythia→RWKV, (iii) an intervention experiment within the same mapping that measures the adapter’s non-linearity for both RWKV→Pythia and Pythia→RWKV, and (iv) measurements of the accuracy, memory, domain generalization, and path asymmetry of single-boundary chimeras in RWKV→Pythia and Pythia→RWKV.

II. Method

a) Residual pairs.: We feed the same text to the frozen A, B , which share a tokenizer, and pair the outputs $r_A^j[t]$ and $r_B^j[t]$ at position t and block j . Residuals are captured directly via forward hooks on each block, and both families and all 32 layers (including layer 0) are treated as quantities of the same kind (see Appendix A for the capture implementation and consistency checks).

b) Shared-latent adapter.: We place one encoder–decoder pair per model and map into a single latent z shared across all layers:

$$\begin{aligned} z_A &= \text{LN}(E_A(r_A)), & z_B &= \text{LN}(E_B(r_B)), \\ \mathcal{L} &= \|D_A(z_A + \xi_A) - r_A\|^2 + \|D_B(z_B + \xi_B) - r_B\|^2 \\ &\quad + \lambda \|z_A - z_B\|^2. \end{aligned}$$

$\xi_A, \xi_B \sim \mathcal{N}(0, \sigma^2 I)$ (sampled independently per branch), $\lambda=1$, LN is non-affine, and σ is set to 0.4. E, D are

shallow residual MLPs. Neither cross mapping $A \rightarrow B := D_B(z_A) = D_B(\text{LN}(E_A(r_A)))$ nor $B \rightarrow A := D_A(z_B) = D_A(\text{LN}(E_B(r_B)))$ is given any cross supervised target (r_A never appears in B 's reconstruction loss, nor r_B in A 's), yet the setup is not “unsupervised” either. Since the hypothesis is that the spaces align under a simple transformation, a high cross read-out obtained with a heavy, highly expressive adapter would not be evidence of a shared space. We therefore restrict the mapping depth to a single residual block, and report the parameter count together with linear controls for RWKV $\rightarrow$ Pythia/Pythia $\rightarrow$ RWKV (§V).

c) Adapter architecture.: E_A, E_B, D_A, D_B share an identical form: a linear projection to the latent width, b pre-norm residual blocks, and a linear projection to the output. The reported configuration is $z=d=4096$ (no compression), $b=1$, and internal width $h=4096$. The second matrix of each residual block is zero-initialized, so each block is initially the identity map and the adapter as a whole starts learning from a linear map. This, however, is a property of the initialization, not of the trained model: as measured in §V, after convergence 13.4%/12.0% of the output variance of RWKV $\rightarrow$ Pythia/Pythia $\rightarrow$ RWKV is nonlinear, and the same nonlinear branch serves both cross read-outs. Against the parents’ combined 14.2B parameters, the adapter has 268.5M (1.9%). “Lightweight” refers not to absolute smallness but to the size relative to the parents and to the low depth.

d) Two design choices that matter.: The non-affine LN prevents the alignment term from being satisfied by shrinking z , and the noise ξ is intended to suppress escape into a token-independent common mode — since a constant cannot evade the noise, z is forced to carry per-token information. We use a fixed $\sigma=0.4$ and do not sweep σ in this paper.

e) Names of the four paths and the chimera.: In the equations we set A to RWKV and B to Tulu-Pythia, but the main text and figures avoid the two-letter codes and state the prefix and suffix explicitly. The four paths are RWKV self-recon, RWKV $\rightarrow$ Pythia, Pythia self-recon, and Pythia $\rightarrow$ RWKV. RWKV $\rightarrow$ Pythia and Pythia $\rightarrow$ RWKV are cross-family chimeras; RWKV self-recon and Pythia self-recon are same-family reconstruction controls that pass through the same boundary and adapter. The switch layer L is the index (starting from 0) of the last block the prefix parent executes, not the number of layers executed — since block 0 is included, the prefix comprises $L+1$ blocks. In RWKV $\rightarrow$ Pythia, we run the first $L+1$ blocks of A (blocks 0 through L) and translate the residual into B space exactly once: $h_B = \text{unstd}_B(D_B(\text{LN}(E_A(\text{std}_A(h_A)))))$. We then run the remaining $31-L$ blocks of B (from $L+1$ onward) and the language-model head. Pythia $\rightarrow$ RWKV is evaluated separately with the same trained adapter, inserting $h_A = \text{unstd}_A(D_A(\text{LN}(E_B(\text{std}_B(h_B)))))$ into the input of RWKV block $L+1$. In both cases the total is 32 blocks,

split as $L+1$ and $31-L$, with a single translation. Correctness of the wiring is guaranteed by the boundary gate — re-insertion of the suffix parent’s own residual at the corresponding layer must recover the parent logits — and we confirmed $\text{rel}=0$ at every boundary (see Appendix A for details of the check).

f) Evaluation metrics.: We measure the coefficient of determination R^2 on text not seen during training (pooling all tokens and all dimensions within a layer, with the SST centered per dimension) for both the cross-family mappings ($A \rightarrow B, B \rightarrow A$) and the same-family mappings ($A \rightarrow A, B \rightarrow B$). Latent alignment is measured with the centered correlation ρ_{ctr} (with the common mode removed), and we additionally report the fraction f of the per-token varying component. The two are linked to the alignment term through the relation $2f(1-\rho_{\text{ctr}})$. Each metric is computed per layer over all tokens (both the SST of R^2 and the common-mode removal for ρ_{ctr} and f are done within each layer), and we report the simple mean over the 32 layers. We never aggregate in one pool across layers — differences among the per-layer means would inflate the apparent correlation and variance. For the chimeras, we give English prompts verbatim and evaluate with generation samples produced by deterministic argmax at temperature 0 (at most 40 new tokens, early stopping only on EOS output), multiple-choice accuracy, and next-token cross-entropy.

III. A Learnable Shared Representation Exists Between Corresponding Layers (Under Favorable Conditions)

For the pair of 7B instruction-following models RWKV-4-Raven-7B (pure RNN) $\leftrightarrow$ Tulu-Pythia-6.9b (Transformer), both with 32 blocks, $d=4096$, and a shared GPT-NeoX tokenizer, aligning on instruction-formatted text (Alpaca) lets an adapter consisting of 1 residual block reach a centered latent correlation of $\rho_{\text{ctr}} = 0.901$ on unseen text. Because the raw correlation (0.97) can be inflated by the common mode, we do not take it as evidence by itself. We take the centered correlation of 0.901, together with the fact that each decoder can reconstruct residuals of its own family ($A \rightarrow A$ 0.70, $B \rightarrow B$ 0.71), as the grounds for a shared representation. Moreover, even the best linear map fitted directly to each cross read-out task reaches $A \rightarrow B$ 0.487 and $B \rightarrow A$ 0.535 (§V). Hence a large part of the observed predictability cannot be explained by the expressive power of the deep nonlinear map alone. However, same-index layer pairs are not intrinsically special: comparing raw residuals, without any adapter, across all 32×32 layer pairs (Fig. 2), linear CKA is weak throughout (diagonal mean 0.05 vs. off-diagonal mean 0.03), and layer j of A is most similar to layer j of B in only 2 of 32 rows (the most similar layer is layer 0 in 14 rows and the final layer in 12 rows). Nor are same-index layers special in direct linear regressability: fitting per-layer linear maps to the same full set of layer pairs yields fit R^2 with a diagonal mean of 0.87 vs. an

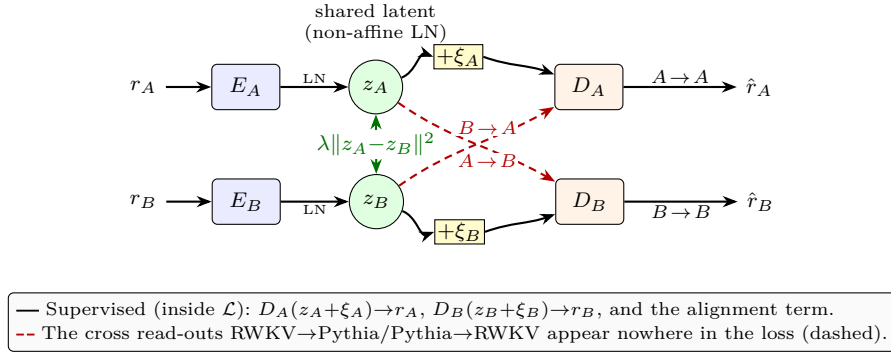


Fig. 1. Shared-latent adapter. The residual of each frozen model is encoded into a single latent z (non-affine LayerNorm) and decoded back to its own family (solid lines; these two reconstructions and the alignment term $\lambda \|z_A - z_B\|^2$ constitute the entire loss). Gaussian noise $\xi_A, \xi_B \sim \mathcal{N}(0, \sigma^2)$ (independent) is added to z only immediately before decoding and does not enter the alignment term. The solid $A \rightarrow A$ (RWKV self-recon) and $B \rightarrow B$ (Pythia self-recon) are same-family reconstructions; the dashed $A \rightarrow B$ (RWKV $\rightarrow$ Pythia) and $B \rightarrow A$ (Pythia $\rightarrow$ RWKV) are cross read-outs. The latter two are never fitted to a cross target and become possible only once the alignment term makes $z_A \approx z_B$.

off-diagonal mean of 0.84, and the same-numbered layer is best in only 3 of 32 rows. Hence $\rho_{\text{ctr}} = 0.901$ is an alignment learned and recovered on the chosen same-index layer pairs, not a correspondence intrinsic to matching layer indices — it should be read as the choice of same-index layers laid on top of a regressability that exists for almost any layer pair.

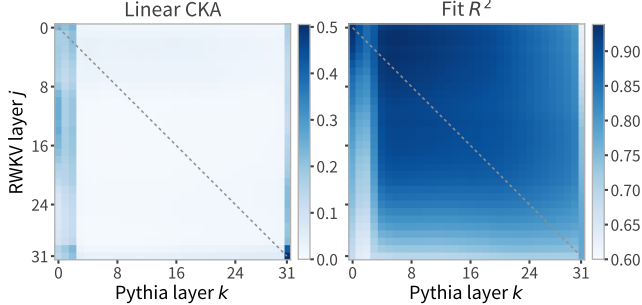


Fig. 2. All 32×32 layer correspondences of raw residuals (generated from the canonical table paper_layer_correspondence.csv). Left: linear CKA is weak throughout, and high values concentrate in the row and column of layer 0 and the corner of the final layer — the diagonal (dashed) is not special. Right: the fit R^2 of per-layer linear maps fitted to the same pairs is high for nearly all pairs, at 0.62–0.94. The color scales of the two panels differ.

We can confirm that the alignment is learnable, but in this sense as well its scope is limited (§IX).

All numbers below come from a single training run of the same configuration. The latent width is $z=d=4096$ (no compression), all 32 blocks are paired, the adapter has 268.5M parameters, $\sigma=0.4$, and the learning rate was decayed from 10^{-3} by a plateau schedule down to a floor of 10^{-5} . The training signal consists of residuals taken from 2,075 windows of the Alpaca training split (each 112 tokens; $\approx 232\text{k}$ independent token positions), forming 7.44M pairs across the 32 layers, and this pair set was iterated over about 228 times. Reported values are from

the checkpoint at the deepest point (415,000 iterations) where held-out $A \rightarrow B$ is best (training runs and convergence records are in Appendix A). The Pythia $\rightarrow$ RWKV cross-family map attains $B \rightarrow A$ 0.590, slightly above $A \rightarrow B$ 0.559 (residuals on the RNN side are easier to recover from the Transformer latent). Both fall below same-family reconstruction ($A \rightarrow A$ 0.695, $B \rightarrow B$ 0.711), corroborating that cross-family transfer is bottlenecked by decoding rather than by alignment. That these alignments and read-outs are a true per-token correspondence rather than distributional overlap was verified with a negative control: keeping the row order of A fixed while re-permuting the residual rows of B under a single random permutation collapses ρ_{ctr} from 0.901 $\rightarrow$ 0.000, $A \rightarrow B$ from 0.561 $\rightarrow$ -0.592 , and $B \rightarrow A$ from 0.592 $\rightarrow$ -0.609 (the same-family maps $A \rightarrow A$, $B \rightarrow B$ are unchanged by construction). That is, neither direction of read-out is obtainable without the correct token correspondence.

a) The common mode is not a trivial issue.: Introducing the fraction of the latent that varies per token, $f = \|z - \bar{z}\|^2 / \|z\|^2$, quantifies this pitfall. Under layer normalization without elementwise gains, the raw correlation decomposes as $\rho_{\text{raw}} = (1 - f) + f \rho_{\text{ctr}}$, i.e., $\text{align} = 2f(1 - \rho_{\text{ctr}})$. The f reported here is the mean of the varying fractions of A and B respectively, and this form holds as an equality when the two are equal and their common modes are parallel. As measured, at evaluation points after training has progressed this relation holds to a precision of about 3×10^{-4} . It does not hold early in training, however: at iteration 0 the residual reaches 6.6×10^{-2} . What the equality tells us is that the raw correlation is a mixture — of which a fraction $1 - f$ is given for free as a constant term carrying no information. Since f at the deepest point is 0.350, 65% of the latent’s energy is a token-independent constant, and the raw correlation is dominated by that constant. Indeed, whereas the centered ρ_{ctr} is 0.901, the raw correlation

reads 0.965, and the equality reproduces this exactly as $(1-0.350) + 0.350 \times 0.901 = 0.965$. The interpretive limits of f (a large f does not necessarily carry information) and a concrete example of training dynamics in which the raw correlation fails to capture improvement are given in Appendix B.

IV. Both Cross Read-Outs Are Constrained by Same-Family Reconstruction

Fig. 3 shows the five measured diagnostic values.

$$\begin{aligned} \rho_{\text{ctr}} &= 0.901, \\ R^2(B \rightarrow B) &= 0.711, \quad R^2(A \rightarrow B) = 0.559, \\ R^2(A \rightarrow A) &= 0.695, \quad R^2(B \rightarrow A) = 0.590. \end{aligned}$$

ρ_{ctr} is a correlation coefficient, whereas the latter four are coefficients of determination; their magnitudes cannot be compared as if on the same scale. The conclusions of this section rest on the facts that, among the R^2 values, $A \rightarrow B \leq B \rightarrow B$ and $B \rightarrow A \leq A \rightarrow A$ hold in all 32 layers, and that in the autoencoder contrast of §VII the decode-only loss also manifests in downstream performance.

High latent correlation does not automatically guarantee high reconstruction accuracy. RWKV→Pythia can be compared with Pythia self-recon, which passes through the same decoder D_B , and Pythia→RWKV with RWKV self-recon, which passes through the same decoder D_A ; the only difference is whether the input is a same-family or a cross-family latent. Both inequalities are empirical observations rather than theorems, but in either direction a substantial error is already present at the self-reconstruction stage, and improving only the cross-family step leaves that error in place.

a) Compression cannot explain it, but the cause of the ceiling remains unresolved.: One could object that $B \rightarrow B$ is low simply because the latent compresses information. This can be checked not by argument but by removing the compression itself. At $z=d=4096$ the composite map $D \circ E$ can represent the identity map, and no rank constraint exists. Even so, $B \rightarrow B$ remains at 0.711. Is it, then, a ceiling set by the noise σ ? A simple scalar approximation cannot explain 0.711, and what is rate-limiting cannot be determined until σ is swept, so we leave this unresolved in this paper. Note that σ plays another role—suppressing collapse into the common mode—which the trajectory of f supports, and that point is unchanged: removing compression alone does not resolve the problem; even with the wide latent, 98% of the latent energy collapses to a constant early in training, and recovery is slow until the final phase of training (the trajectory of f is given in Appendix B). This is a degeneracy that persists even when a wide latent is used, and it is also the reason the common-mode correction is not a trivial issue.

V. The Adapter Is Nonlinear on Both Cross Paths

The adapter is a single zero-initialized residual block sandwiched between two linear maps, so training begins

from a purely linear map. But that is a property at iteration 0, not after 415,000 iterations. We measured how linear the adapter at the deepest point actually is, for both RWKV→Pythia and Pythia→RWKV, on the same 5 representative layers and the same fit and held-out rows.

How much of the adapter is linear. The best affine map mimicking the adapter’s own output explains 86.6% of the output variance for RWKV→Pythia and 88.0% for Pythia→RWKV. Hence the remaining 13.4%/12.0% cannot be expressed by any affine map. The fact that the second matrix is zero-initialized speaks only to initialization; the trained map is clearly nonlinear on both paths.

How far a linear map alone can reach. The comparison is not an approximation of the adapter: it is a control obtained by directly fitting, per layer and separately for each path, the best affine map from the standardized input residuals to the standardized output residuals via ridge least squares (see Appendix A for setup details). Evaluated on the held-out rows of the 5 representative layers {4, 10, 16, 22, 28}, the best direct affine map reaches 0.487 for RWKV→Pythia and 0.535 for Pythia→RWKV. The corresponding adapter, on the same rows, achieves 0.558 for RWKV→Pythia and 0.576 for Pythia→RWKV, so the nonlinear map’s increment is +0.071/+0.041. Because even a low-capacity affine map achieves substantial cross-family prediction on both paths, the shared-representation claim is underwritten by this linear reachability.

The separately fitted affine control. In Table III, replacing the RWKV→Pythia adapter with the direct affine map collapses accuracy by 8–23 points. For Pythia→RWKV, by contrast, the gap is small, and on SciQ @8 the affine control is 3.4 points higher. This matters as a path difference, but because the two maps differ in training objective, data, and regularization, we do not treat it as a causal effect of nonlinearity.

A confound-controlled intervention: disabling the MLP branch of the same adapter. We therefore scale only the residual block of the same trained adapter (its sole learned nonlinearity, $\text{fc}_2(\text{GELU}(\text{fc}_1(\cdot)))$) by a coefficient α ($\alpha=1$ intact; $\alpha=0$ is the identical map with the branch removed). Sweeping representation R^2 over $\alpha = 0, .25, .5, .75, 1$ yields $-0.41, -0.75, 0.35, 0.52, 0.56$ for RWKV→Pythia and $-0.30, -0.66, 0.36, 0.53, 0.58$ for Pythia→RWKV. On both paths, removing the branch collapses performance all the way to negative R^2 , and moreover the sweep is non-monotonic. The two linear layers were trained in coordination with the residual block, and in this trained adapter, the nonlinear branch cannot be removed from the read-out of Pythia→RWKV any more than from RWKV→Pythia. This is a property of the trained map and does not mean the task is in principle unsolvable by a linear map (we did not compare against a linear adapter trained with the same objective). For multiple choice as well, we evaluate the same $\alpha = 0/1$ pair on identical questions in paired fashion.

The branch effect in Table I is +11.1–+39.6

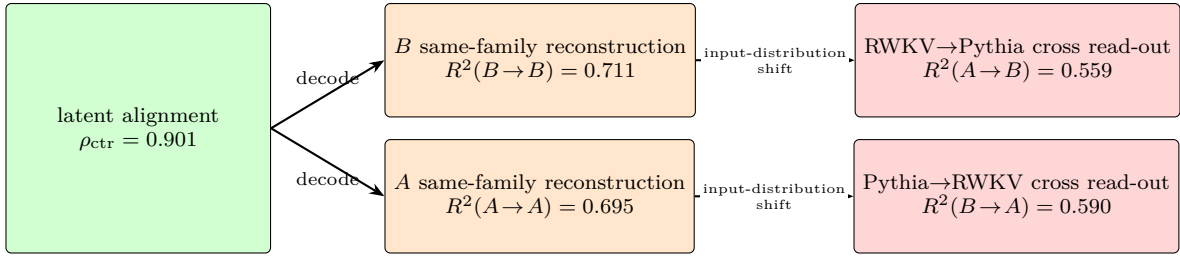


Fig. 3. Diagnostic values from the shared latent to the read-outs in both directions. High centered correlation coexists with imperfect self-reconstruction, and the RWKV→Pythia/Pythia→RWKV cross read-outs fall below, in all layers, the Pythia self-recon/RWKV self-recon self-reconstructions that use the same respective decoders. Because a correlation coefficient and a coefficient of determination are different scales, we do not interpret the numeric gap between the scales itself as an amount of loss.

TABLE I

Length-normalized accuracy (%) when the nonlinear branch of the same adapter is disabled. Each cell shows $\alpha : 1 \rightarrow 0$ (full trained map $\rightarrow$ the same map with only the branch removed). ARC-Easy 2376, SciQ 1000, chance 25%.

L	ARC-Easy		SciQ	
	RWKV→Pythia	Pythia→RWKV	RWKV→Pythia	Pythia→RWKV
4	59.6 → 31.9	62.2 → 31.8	70.9 → 31.3	69.2 → 32.6
8	59.0 → 29.9	57.3 → 35.5	67.7 → 31.4	64.6 → 37.0
16	48.6 → 28.5	57.5 → 43.7	54.5 → 31.7	68.0 → 47.3
24	47.0 → 34.8	59.6 → 48.4	51.1 → 40.0	69.9 → 49.1

points for RWKV→Pythia and +11.2+36.6 points for Pythia→RWKV, and an exact McNemar test on identical questions gives $p < 2 \times 10^{-9}$ in all 16 comparisons. Hence the necessity of the nonlinear branch is not inferred from one path alone: it is confirmed directly in both the RWKV→Pythia and Pythia→RWKV representations and in downstream accuracy.

VI. Syntactically Plausible Generation, Degraded Factuality

Here we actually compose both cross-family paths at 7B. At switch layer L , we run the prefix parent from block 0 through L (a total of $L+1$ blocks), translate its residual once into the other space via the boundary formula of §II, and then run the suffix parent’s blocks from $L+1$ onward together with its language-model head—not a per-block round trip. Thus RWKV→Pythia ($L=16$) uses 17 RWKV blocks and 15 Pythia blocks, and Pythia→RWKV ($L=16$) the reverse; neither is an exact half.

We insert the same deepest-point adapter at $L=4$ for both RWKV→Pythia and Pythia→RWKV, and greedily decode the same four English prompts at temperature 0 with at most 40 new tokens. $L=4$ is the boundary at which the two-task average is largest among the four boundaries in each cross-family direction, and it is the common boundary at which both directions enter the non-dominated set on the SciQ KV–accuracy plane (Fig. 5). This is, however, a representative-point selection made after seeing the same reported tasks, not an independent selection criterion for generation. This comparison is not an exhaustive generation benchmark, but at least it does

not infer one direction’s generation from the other’s cross read-out; we observe both end-to-end wirings under identical conditions.

Two points can be observed from these examples. First, both RWKV→Pythia and Pythia→RWKV can generate syntactically plausible output. Both answer the capital of France correctly, and both start the primary colors correctly. RWKV→Pythia also answers the black-hole prompt correctly in one sentence, so this is not behavior inferred from the representation metric of only one direction. Second, fluency can be preserved while content or instruction format is lost. Both directions complete the water prompt incorrectly with “hydrogen”, and Pythia→RWKV drifts into a dialogue format after correctly answering the primary colors and explains the black hole circularly. Thus even cross read-outs of 0.559/0.590 are not sufficient for factuality. A more faithful chimera requires improving both cross read-outs and same-family reconstruction.

VII. Function and Inference-Memory Comparison Against the Parents

Assessing the practical value of NinaXander requires a functional comparison against the frozen parents, not internal-representation metrics alone. Here we measure (i) how much performance can be retained relative to the strong parent, (ii) whether the weak parent or a Transformer with the same KV capacity is surpassed, and (iii) how much memory is saved in exchange.

This comparison is run in a multiple-choice format that computes the log-likelihood of each option and selects the maximum. Because no generation, string matching, or subjective judgment is involved, the result is uniquely determined. We used the full evaluation sets (ARC-Easy test 2376, SciQ test 1000), the $z=4096$ adapters at the deepest point (415,000 iterations), and four switch positions. Between-configuration comparisons on the same question set are paired data, so differences are assessed with paired tests (McNemar’s exact test and paired bootstrap) rather than independent per-cell binomial SEs. Because the options differ in length and sum vs. length-normalized scoring can disagree in ranking, both are stored in the structured data, and the main text uses length-normalized

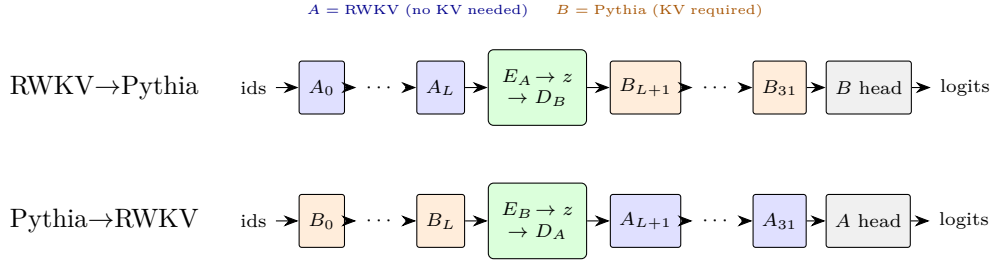


Fig. 4. Symmetric construction of the RWKV→Pythia/Pythia→RWKV chimeras. The parent on the left of the figure runs up to block L ; its residual is translated once and passed to the right parent’s blocks from $L+1$ onward and its head. Blue is RWKV-Raven, orange is Tulu-Pythia. The number of Transformer layers is therefore $31-L$ for RWKV→Pythia and $L+1$ for Pythia→RWKV.

TABLE II

Regeneration results of the deepest-point adapter via RWKV→Pythia/Pythia→RWKV (both $L=4$), together with the outputs of the two parents, pure Pythia and pure RWKV, on the same prompts. The English originals were fed verbatim, using temperature 0 argmax, at most 40 new tokens, and early stopping on EOS only. Each chimera is placed next to its suffix parent. Long repetitions are summarized; the verbatim outputs are recorded in paper `cross_family_generation_samples.csv`. This is a qualitative comparison limited to four prompts, not a quantitative evaluation of generation quality.

Prompt (verbatim)	pure Pythia	RWKV→Pythia cross read-out $R^2=0.559$	pure RWKV	Pythia→RWKV cross read-out $R^2=0.590$
### Instruction: What is the capital of France?	“The capital of France is Paris.”	“The capital of France is Paris.” ✓	“Paris.”	“Paris.” ✓
### Response: The three primary colors are	“red, yellow, and blue. These colors are often used in combination to create a wide variety of colors.”	“red, yellow, and blue.” ✓	Correct with “red, blue, and yellow.”, then drifts into a Q&A asking for the opposite color of red (green)	Correct with “red, blue, and yellow.”, then drifts into a Bob/Alice dialogue about green
Question: Water is made of hydrogen and Answer:	“Hydrogen is made of hydrogen and oxygen.”	“Hydrogen.” (incorrect)	“hydrogen.” (incorrect)	“hydrogen.” (incorrect)
### Instruction: Explain what a black hole is in one sentence. ### Response:	“A black hole is a region of spacetime where gravity is so strong that nothing, not even light, can escape its pull.”	“A black hole is a region in space where gravity is so strong that nothing, not even light, can escape its pull.” ✓	“A black hole is a region of space where the gravitational pull is so strong that nothing, not even light, can escape.”	“A black hole is a type of celestial object that is formed when a star collapses and collapses into a black hole.” (circular)

scoring for reporting and testing. Moreover, SciQ is evaluated in a closed-book format (Question/Answer only) that does not supply the accompanying support passages, so the SciQ values in this paper are not directly comparable to published values under the standard protocol that provides support.

Six observations follow from the joint comparison of the four paths. (a) Even with half of the 7B being foreign, both RWKV→Pythia and Pythia→RWKV can actually answer the questions. All cross configurations far exceed chance, reaching RWKV→Pythia 70.9% and Pythia→RWKV 69.9% on SciQ. (b) The depth effect is path-dependent. RWKV→Pythia decreases monotonically over $L = 4 \rightarrow 24$, SciQ $70.9 \rightarrow 67.7 \rightarrow 54.5 \rightarrow 51.1$ and ARC-Easy $59.6 \rightarrow 59.0 \rightarrow 48.6 \rightarrow 47.0$, whereas Pythia→RWKV is non-monotonic, SciQ $69.2 \rightarrow 64.6 \rightarrow 68.0 \rightarrow 69.9$ and ARC-Easy $62.2 \rightarrow 57.3 \rightarrow 57.5 \rightarrow 59.6$. Local read-out at the same four layers is RWKV→Pythia 0.570/0.555/0.575/0.567 and

TABLE III

Length-normalized multiple-choice accuracy (%), ARC-Easy 2376, SciQ 1000, chance 25) for the trained RWKV→Pythia/Pythia→RWKV adapters and the best affine controls fitted separately to each path’s input and output residuals. Each value uses the same questions, the same switch layer, and the same suffix execution.

L	ARC-Easy				SciQ			
	RWKV→Pythia adapter	Pythia→RWKV affine	Pythia→RWKV adapter	Pythia→RWKV affine	RWKV→Pythia adapter	Pythia→RWKV affine	Pythia→RWKV adapter	Pythia→RWKV affine
4	59.6	41.9	62.2	60.2	70.9	47.5	69.2	69.6
8	59.0	43.1	57.3	56.0	67.7	49.1	64.6	68.0
16	48.6	34.8	57.5	57.5	54.5	35.9	68.0	69.6
24	47.0	39.4	59.6	59.2	51.1	42.7	69.9	68.2

Pythia→RWKV 0.552/0.517/0.604/0.608, so the end-to-end ranking is not determined by the layer index or a single local R^2 alone. (c) The stronger parent is not surpassed on either path. All 8 cross configurations fall below Tulu-Pythia on both tasks, and the differences are significant

TABLE IV

Length-normalized accuracy (%) of the four paths. RWKV self-recon/Pythia self-recon are same-family reconstruction controls; RWKV→Pythia/Pythia→RWKV are cross-family chimeras. Differences between the two cross paths on the same questions and switch layer were paired and tested.

L	RWKV self-recon		RWKV→Pythia		Pythia self-recon		Pythia→RWKV	
	ARC	SciQ	ARC	SciQ	ARC	SciQ	ARC	SciQ
4	62.8	68.5	59.6	70.9	62.2	74.8	62.2	69.2
8	59.8	66.7	59.0	67.7	62.7	74.8	57.3	64.6
16	54.8	59.8	48.6	54.5	57.4	71.1	57.5	68.0
24	52.2	57.9	47.0	51.1	59.4	70.1	59.6	69.9

under McNemar tests on the same questions in every case. (d) The difference against RWKV is likewise path- and task-dependent. RWKV→Pythia ($L=4$) exceeds RWKV by +3.9 points on SciQ ($p=0.018$) but is −2.2 points on ARC-Easy ($p=0.042$). Under sum scoring, however, this advantage disappears (RWKV→Pythia ($L=4$) is −2.7 points on SciQ; Pythia→RWKV ($L=4$) is −4.6 points on ARC / −5.7 points on SciQ) — the verdict against the weaker parent depends on the choice of length normalization. Pythia→RWKV does not significantly exceed RWKV at any boundary, and Pythia→RWKV ($L=4$) is statistically equivalent at ARC +0.4 points ($p=0.66$) and SciQ +2.2 points ($p=0.14$). (e) The difference against the separately fitted affine controls is also path-dependent. On RWKV→Pythia the adapter is 8–23 points higher in every cell, whereas on Pythia→RWKV the differences are small, and at SciQ @8 the affine is 3.4 points higher. This comparison is descriptive because the mappings themselves differ; the causal comparison that manipulates the nonlinear branch of the same adapter is carried out for both paths via the α intervention in §V. (f) The gap to the same-family controls is not translation error alone. With the B suffix, accuracy decreases in the order pure Pythia, Pythia self-recon, RWKV→Pythia, with Pythia self-recon–RWKV→Pythia gaps of SciQ 4–19 points and ARC 3–12 points. With the A suffix, by contrast, Pythia→RWKV conversely exceeds RWKV self-recon at deep boundaries, and the differences at $L = 16/24$ are ARC +2.7/+7.4 points and SciQ +8.2/+12.0 points. The same-family controls therefore isolate the decode stage, but the gap to the cross paths contains both the prefix parent’s capability and translation mismatch, and does not constitute a universal “translation cost.”

a) Depth dependence differs between RWKV→Pythia and Pythia→RWKV.: As Table IV shows, at $L=16, 24$ Pythia→RWKV exceeds the same-layer RWKV→Pythia by +8.9/+12.6 points on ARC and +13.5/+18.8 points on SciQ (McNemar tests on the same questions, all $p < 3 \times 10^{-15}$). At $L=4$, on the other hand, the difference is +2.6 points on ARC ($p=0.015$) but −1.7 points on SciQ ($p=0.28$), so at shallow boundaries the path difference is not uniform. Hence the monotonic depth effect seen for RWKV→Pythia cannot be generalized as “the difficulty of joining

layers with the same index.” Every Pythia→RWKV configuration falls significantly below Pythia. Nor is there any Pythia→RWKV configuration that significantly exceeds RWKV; Pythia→RWKV ($L=4$) is equivalent at ARC 62.2 vs. 61.8 ($p=0.66$) and SciQ 69.2 vs. 67.0 ($p=0.14$).

The best affine mapping for Pythia→RWKV is a control fitted separately from the adapter at each switch layer. At ARC @4 the adapter is +2.0 points higher ($p=0.005$), but most of the other cells show no difference, and at SciQ @8 the affine is conversely +3.4 points higher ($p=0.003$). Separately from this between-mapping comparison, the intervention that sets the nonlinear branch of the same adapter to $\alpha = 0$ is executed for RWKV→Pythia/Pythia→RWKV at the same four layers and the same questions, and the causal contribution of the nonlinear branch is assessed from those paired results. The comparison with the $A \rightarrow A$ autoencoder likewise replaces the RWKV prefix simultaneously with a Pythia prefix and cross translation, and thus does not isolate translation error alone.

b) The released defaults are “best measured” selections, not external validation.: Averaging the accuracies of the two tasks, Pythia→RWKV ($L=4$) is 65.70% and RWKV→Pythia ($L=4$) is 65.25%, a difference of only 0.45 points. The full release splits by path explicitly: ninaxander-tulu69-to-raven7b-best sets Pythia→RWKV ($L=4$), and ninaxander-raven7b-to-tulu69-best sets RWKV→Pythia ($L=4$), as the default within each path. The runtime refuses switching to the path opposite to the package name, while retaining the $L=8, 16, 24$ plans for boundary comparison. Both are post-hoc rankings selected from each path’s four boundaries on the same two reported test sets, and there is no independent deployment-selection split. They are therefore not used as grounds for claiming improvement on unseen execution paths in general or superior generalization performance.

c) The memory effect of composition: the KV cache disappears together with the Transformer layers.: Performance and inference memory must be evaluated jointly, because the two families do not have the same inference-time cost. Transformer layers cache key/value per token, whereas an RNN layer carries only a recurrent state that is $O(1)$ in sequence length. RWKV→Pythia retains only $31-L$ Transformer layers, and Pythia→RWKV only $L+1$. The Tulu-Pythia cache was measured, not assumed: 16.0 KiB/token per layer, 512 KiB/token over 32 layers (in exact agreement with $2Ld \cdot 2$ bytes).

Including RWKV→Pythia/Pythia→RWKV, Pythia→RWKV ($L=4$) reduces the Transformer layers from $27 \rightarrow 5$ relative to RWKV→Pythia ($L=4$) while nearly preserving the two-task average, 65.25% $\rightarrow$ 65.70%. The recurrent state of the replacing RWKV is 40 KiB per layer (a computed value from 5 state vectors $\times$ 4096 dimensions $\times$ fp16) and is by construction independent of context length. However, pure RWKV has zero KV

TABLE V

Accuracy and inference memory for both RWKV→Pythia/Pythia→RWKV paths. KV is measured; the RWKV recurrent state is constant in context length.

Configuration	Transformer layers	KV / token	KV @ 4k context	Reduction	ARC / SciQ
pure Tulu-Pythia	32	512 KiB	2.00 GiB	—	65.3 / 81.0
RWKV→Pythia ($L=4$)	27	432 KiB	1.69 GiB	15.6%	59.6 / 70.9
RWKV→Pythia ($L=8$)	23	368 KiB	1.44 GiB	28.1%	59.0 / 67.7
RWKV→Pythia ($L=16$)	15	240 KiB	0.94 GiB	53.1%	48.6 / 54.5
RWKV→Pythia ($L=24$)	7	112 KiB	0.44 GiB	78.1%	47.0 / 51.1
Pythia→RWKV ($L=24$)	25	400 KiB	1.56 GiB	21.9%	59.6 / 69.9
Pythia→RWKV ($L=16$)	17	272 KiB	1.06 GiB	46.9%	57.5 / 68.0
Pythia→RWKV ($L=8$)	9	144 KiB	0.56 GiB	71.9%	57.3 / 64.6
Pythia→RWKV ($L=4$)	5	80 KiB	0.31 GiB	84.4%	62.2 / 69.2
pure RWKV-Raven	0	0	0	100%	61.8 / 67.0

and the ranking is task-dependent, so Fig. 5 is presented as a memory–accuracy frontier restricted to SciQ.

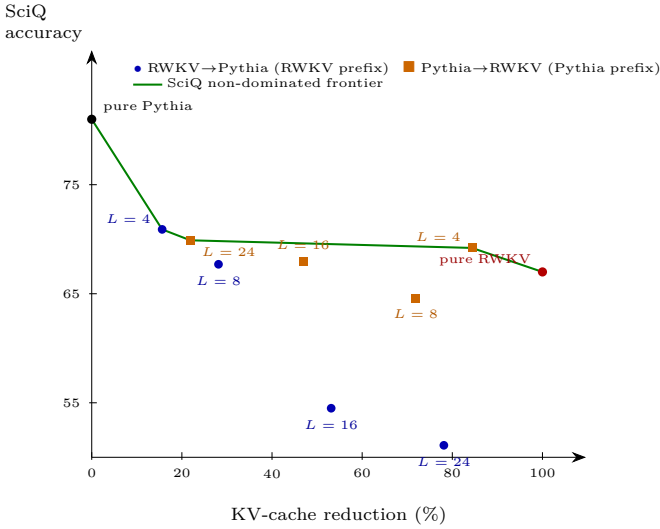


Fig. 5. SciQ accuracy vs. measured KV reduction for RWKV→Pythia/Pythia→RWKV. The non-dominated points across both paths are pure Pythia → RWKV→Pythia ($L=4$) → Pythia→RWKV ($L=24$) → Pythia→RWKV ($L=4$) → pure RWKV (green). This is a post-hoc frontier on the single task SciQ, and does not imply path-general superiority or generalization to unseen tasks.

Adding the Pythia-prefix path yields the high-reduction point Pythia→RWKV ($L=4$). This point does not appear with the RWKV-prefix path alone. Meanwhile, pure RWKV still has zero KV, and Pythia→RWKV ($L=4$) does not significantly exceed RWKV either. The benefit is therefore the ability to create additional trade-off points between the frozen parents, not that the new configurations uniformly dominate the parents.

d) Equal-KV Transformer baselines and an inference implementation with unused blocks removed.: We measured comparisons against efficiency baselines on the Transformer-parent side. For RWKV→Pythia, compared with a Pythia early exit holding the same $31-L$ Transformer blocks, the chimeras at @16/@24 are higher by ARC +11.2/ + 18.5 and SciQ +12.8/ + 18.7 points, while @4 is tied. For Pythia→RWKV, against

early exits with the same $L+1 = 5/9/17/25$ blocks, the chimera is ARC +35.4/ + 27.6/ + 16.9/ + 2.1 and SciQ +38.7/ + 31.6/ + 17.9/ + 0.5 points. The first three boundaries are significant, and at 25 blocks it is a tie (ARC $p=0.069$, SciQ $p=0.80$): when the Pythia prefix is very shallow, the RWKV suffix supplies real function.

A path that physically removes the unused blocks is also implemented for RWKV→Pythia/Pythia→RWKV. For Pythia→RWKV, the Pythia prefix, the RWKV suffix, and the full path joining the two are each compared against the unpruned reference, with $\text{rel}=0$ at all 4 boundaries. Resident weights are RWKV→Pythia 13.93–14.55 GiB and Pythia→RWKV 13.49–14.11 GiB (the default Pythia→RWKV ($L=4$) is 14.11 GiB), roughly half of the full pair of parents at 27 GiB. However, the corresponding Pythia early exits are even lighter, 2.64–10.14 GiB for Pythia→RWKV. Furthermore, the reference runtime is not an optimized serving engine: it runs HF RWKV sequentially along the sequence, and cacheless decode recomputes the prefix every time. Pythia→RWKV ($L=4$) prefill at context 2048 takes 22.6 seconds even after pruning (pure Pythia 0.378 seconds), and unpruned cacheless decode runs at 0.148 token/s (Pythia 9.49 token/s). These are figures of the current correctness-first implementation, not architecture-inherent speed limits. Comparisons against KV quantization and sliding-window are also not performed.

The comparison against the parents comes with qualifications: function is preserved, but this is not a uniform improvement in accuracy. Both RWKV→Pythia and Pythia→RWKV fall significantly below the strong Pythia. RWKV→Pythia exceeds RWKV only on SciQ, and Pythia→RWKV does not significantly exceed RWKV at any boundary. The main benefit obtained is that Pythia→RWKV ($L=4$) removes 84.4% of the Transformer KV while retaining part of Pythia’s accuracy.

e) Translation is domain-specific.: The accuracies above were measured on short questions close to instruction format. KV-cache savings matter for long contexts, and how well the chimera preserves open-text language modeling there must be measured separately. At context

512, RWKV→Pythia ($L=4$) has perplexity 6.40 on Alpaca and 93.87 on WikiText-103. Pythia→RWKV ($L=4$) improves in both domains at 3.85/45.00, but WikiText still remains far from the parents’ 11.33/16.22. Extending the context to 2048, Pythia→RWKV ($L=4$) reaches Alpaca 3.50 and WikiText 34.78, improving rather than collapsing. The main cause is therefore not context length but domain shift. Pythia→RWKV mitigates the shift, but WikiText 34.78 is 4.2 times the better parent RWKV’s 8.27, and does not resolve the problem.

The linear control at each switch layer is a supervised affine oracle re-estimated separately on the first 40,000 tokens of Alpaca and of WikiText and evaluated on the respective non-overlapping suffix. The WikiText values are not out-of-domain generalization of a mapping estimated on Alpaca, and they also differ from the trained adapter in objective and data. Hence we do not conclude from the adapter-vs-affine comparison that “only the nonlinear part overfit the domain.” The evidence for domain specificity is that the same trained adapter changes its performance relative to the parents substantially between Alpaca and WikiText.

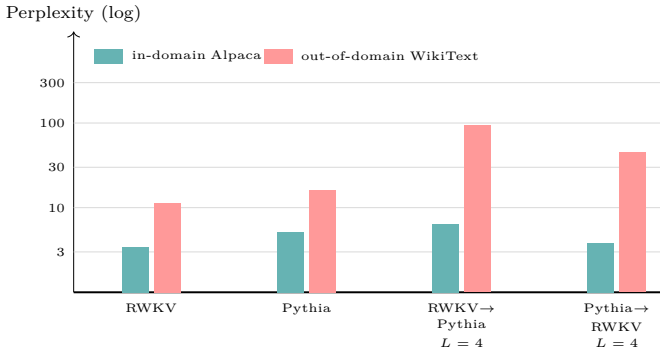


Fig. 6. Domain shift (context 512, log axis). Pythia→RWKV improves the WikiText perplexity of RWKV→Pythia ($L=4$) from 93.87 → 45.00 but does not reach the parents’ 11.33/16.22. Because the same adapter sits near the parents on Alpaca, domain, not length, is the primary difference.

VIII. Related Work

a) Model stitching.: Connecting the layers of 2 frozen models through a trainable stitching layer was introduced by Lenc and Vedaldi [1] to measure the equivalence of representations, and was revisited by Bansal et al. [2] as a probe for representation comparison. Both stitch within the same architecture and the same task; our work differs in that it stitches different families (RNN↔Transformer) through a shared latent that translates each layer only 1 time.

b) Composing frozen LLMs.: Recent work continues to compose frozen LLMs through trainable connections. StitchLLM [3] stitches blocks of Transformers of different scales with stitching layers and selects a path at serving time, and BTS [4] re-merges a set of experts branched

from the same seed back into the seed with stitch layers. CALM [5] fuses the intermediate representations of 2 frozen Transformers via cross-attention, and Manticore [6] fine-tunes end-to-end a hybrid that joins Transformer and Mamba blocks with projections. However, these either have Transformers on both sides (BTS uses a single seed) or fine-tune the composed model (Manticore). To our knowledge, no prior work translates the residuals of a frozen pure RNN and a frozen Transformer bidirectionally, composes them at a single boundary, and goes as far as reporting the measured KV reduction.

c) Representation similarity.: CKA [7] and (SV/PW)CCA [8], [9] are standard tools for measuring representational similarity between networks. Our work uses them as auxiliary metrics (the all-layer correspondence in §III) while placing its main focus on functional, reconstruction-mediated stitching.

d) Model merging and knowledge fusion.: Methods that merge models in weight space—model soups [10], TIES [11], and evolutionary merging [12]—presuppose identical or compatible weights. FuseLLM [13], which fuses output distributions, does not require weight compatibility, but it needs token alignment across vocabularies and continued training of the target model. Our work is complementary in that it merges no weights at all and instead translates the activation space to recombine frozen blocks.

e) Target models.: RWKV [14] is an architecture that has no softmax self-attention and operates as a pure recurrence at inference time; this work uses its fourth-generation instruction-following variant, RWKV-4-Raven-7B. Pythia [15] and its Tulu SFT variant are Transformers. The fact that the two share the GPT-NeoX tokenizer is what makes position-aligned residual pairs possible.

IX. Discussion

In our experiments, a shared representation can be learned between corresponding layers of frozen heterogeneous families ($\rho_{ctr} = 0.90$). However, the scope of this claim is limited: the correspondence in this work is measured under the favorable conditions of an identical tokenizer, identical width, identical layer count, identical inputs, and a domain close to Alpaca; the shuffled control rules out token-independent distribution matching, but we do not disentangle whether the observed correspondence stems from semantics, vocabulary, position, or syntax. Indeed, the substantial degradation of translation on out-of-domain WikiText (§VII) indicates that this may be a representation transformation specific to the training domain. Hence it is more accurate to read this not as a “general shared semantic space” but as a “shared space of token-aligned representations under favorable conditions.” It is also important that high latent correlation does not guarantee high read-out accuracy; the presence of a shared representation and decodability must be evaluated separately. A chimera built on this space functions, but

at a loss, and in the generation samples both paths remain fluent while erring in content or form. Moreover, the fact that Pythia→RWKV substantially outperforms RWKV→Pythia at deep boundaries shows that local R^2 or layer index alone cannot predict end-to-end performance, and that which parent is placed as the prefix matters. A competitive chimera requires identifying the factors that bottleneck $B \rightarrow B$ (including a σ sweep), training across domains, and selecting complementary parent models on the pretraining side.

a) Limitations.: Adapter training uses only a single random seed. The evaluation texts were obtained by splitting the instruction data without overlap, so by construction no training data can leak into them, and the gap between training-time and evaluation-time performance persisting to the end corroborates this; however, at the 7B scale we did not go as far as an n -gram audit. Performance is measured on the full evaluation sets (ARC-Easy 2376, SciQ 1000), and differences between configurations were assessed with paired tests on the same questions (McNemar’s exact test and paired bootstrap). RWKV→Pythia exceeds the weaker parent only on SciQ, and only under the length-normalized score (it does not hold under the sum score), and Pythia→RWKV does not significantly exceed RWKV at any boundary. We evaluate 2 tasks $\times$ 2 cross paths $\times$ 4 switches plus multiple controls, without correction for multiple comparisons. The two direction-fixed released runtimes both take $L=4$ as the default, but this was post-hoc selected from the four boundaries of each path on the same two test sets, with no independent selection/confirmation split. The identical-adapter intervention on the nonlinear branch was performed at the same layers and questions for both RWKV→Pythia/Pythia→RWKV, but it is an intervention on a single checkpoint and may not generalize to different adapter training conditions. Moreover, the composition always has only a single boundary, and we did not measure how translation error accumulates across multiple boundaries. Furthermore, the chimera evaluation is likelihood-based, which measures the ability to select the correct option and does not directly measure the ability to generate. The four generation samples each for RWKV→Pythia/Pythia→RWKV are only a qualitative supplement, not a comprehensive evaluation of generation ability. In addition, the translation is specialized to the training domain and degrades substantially on WikiText. Long-context CE is measured at contexts 512/1024/2048 (capped at 2048 because both parents are models with context length ≤ 2048). This is language-modeling quality and does not directly measure reading comprehension requiring long-range dependencies. Regarding the inference implementation, we measured that actually removing the unused blocks reduces the resident weights by about half (§VII); however, the correctness-first unpruned runtime recomputes the prefix cachelessly, and HF RWKV also runs sequentially, so the chimera is about 63–70 $\times$ slower

than pure Pythia in prefill/decode. This is an implementation property, not a property of the architecture, and latency and throughput on an optimized inference path are left for future work. We also did not perform an equal-memory comparison against KV quantization or sliding-window approaches. Ablations over σ , λ , latent width, the number of residual blocks, and the amount of training data, as well as alignment on a same-family pair (identical architecture), were not conducted in this 32-layer configuration and are left for future work.

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Appendix A

Reproduction Information

- Models: RWKV-4-Raven-7B (BlinkDL .pth converted to HF format and re-saved in fp16) and allenai/open-instruct-pythia-6.9b-tulu (fp16). Both have 32 layers, $d=4096$, and the GPT-NeoX-20B tokenizer.
- Adapter: $z=d=4096$, 1 residual block (inner width 4096, second matrix zero-initialized), non-affine LN, $\sigma=0.4$, $\lambda=1$, 268.5M parameters. AdamW, effective batch 4096 on $4\times V100$, learning rate $10^{-3} \rightarrow 10^{-5}$ ($\times 0.5$ whenever held-out $A \rightarrow B$ stops improving), a 416,000-iteration run (the reported checkpoint is the one at 415,000 iterations, best on held-out $A \rightarrow B$), random seed 0.
- Data: residuals are taken from the leading portion of the Alpaca training split (windows of 112 tokens, $\approx 232k$ independent token positions), and evaluation windows are taken from character position 2,000,000 onward (a fixed offset independent of N); the two do not overlap. Standardization statistics are computed from each layer’s training residuals and used for de-standardization.
- Evaluation: held-out R^2 is computed at each layer by pooling all tokens and all dimensions (SST is centered per dimension; exact metric definitions are

given in Appendix B), and we report the simple mean over the 32 layers. Multiple choice is decided by the log-likelihood of the sequence formed by concatenating each option (sum/mean over the tokens after the prompt), using the full evaluation sets (ARC-Easy test 2376, SciQ test 1000). SciQ is evaluated in closed-book form without the support passage. Differences between configurations are assessed with paired tests (McNemar and paired bootstrap). For the RWKV suffix of Pythia $\rightarrow$ RWKV, options are always evaluated at batch 1; concatenating controls along the batch axis produced per-layer differences of $\text{rel}=8.3\times 10^{-4}$ – 1.8×10^{-3} and was therefore rejected, and only the path that keeps batch 1 while parallelizing over independent CUDA streams is used, after passing a $\text{rel} < 10^{-4}$ gate against sequential execution. Pythia $\rightarrow$ RWKV is likewise fixed to the same distributed checkpoint as RWKV $\rightarrow$ Pythia, step 415,000 selected on held-out $A \rightarrow B$; no re-selection of checkpoints that saw $B \rightarrow A$ or Pythia $\rightarrow$ RWKV QA is performed. However, the two direction-fixed released runtimes take as default the $L=4$ whose reported test mean was highest over the four boundaries of each path after evaluation, with no independent deployment-selection split.

- Code and structured data: the repository for this work is <https://github.com/kotama7/NinaXander> (a private repository as of July 30, 2026; publication planned). It contains the experiment code, the raw data of each evaluation together with aggregate tables of the values reported in the paper, the reproduction procedure, and checks that automatically cross-verify the numbers in the text against the tabulated data. Details of the hook-capture consistency checks, boundary gates, training runs and convergence records, noise-ceiling analysis, and the fitting procedure for the affine control are documented in the repository documentation (docs/).
- Hugging Face distributions: the inference artifacts as of July 30, 2026 were stored in the following three private model repositories. Access requires the owner’s approval, and they are not visible to unauthenticated readers. The private collection grouping the three is <https://huggingface.co/collections/kotama7/ninaxander-private-inference-releases-6a6b02ec40c18d6b41ed040a>.
 - Shared adapter: <https://huggingface.co/kotama7/ninaxander-raven7b-tulu69-adapter>
 - Pythia $\rightarrow$ RWKV (default $L=4$): <https://huggingface.co/kotama7/ninaxander-tulu69-to-raven7b-best>
 - RWKV $\rightarrow$ Pythia (default $L=4$): <https://huggingface.co/kotama7/ninaxander-raven7b-to-tulu69-best>

Each repository includes model cards and notices in

English, Japanese, and Simplified Chinese, SafeTensors, and a SHA-256 manifest. All distribution licenses follow the more restrictive parent (Tulu-Pythia), namely the AI2 AI Model License (non-commercial research only), and each NOTICE records attribution to the parent models.

Appendix B

Metric Definitions and Detailed Common-Mode Analysis

This section gives rigorous definitions of the metrics and supplementary material on the common-mode correction of §III–§IV.

a) Metric definitions.: Residuals are standardized per layer using the per-dimension means and standard deviations estimated from the training tokens (and de-standardized at chimera insertion). All three loss terms are means over all elements (tokens $\times$ dimensions), and since z has unit variance under the non-affine LN, $\lambda=1$ amounts to adding the per-element errors on the same scale. The noise ξ_A, ξ_B is sampled independently for each branch. R^2 is computed per layer as $1 - \text{SSE}/\text{SST}$, pooling the held-out tokens and all 4096 dimensions. SST is the total variation after removing the per-dimension token mean, so dimensions with larger variance contribute more. There are two held-out sets: the training-time evaluation that provides the headline values and the learning curves uses 1,000 windows $\times$ 112 = 112,000 tokens, while the per-layer analysis and the shuffle control (§III) use 400 windows $\times$ 112 = 44,800 tokens; because the evaluation sets differ, the 32-layer averages in the per-layer table (0.561/0.592) do not exactly match the headline values (0.559/0.590). ρ_{ctr} is a single Pearson correlation computed after removing each dimension’s token mean within a layer and then flattening (tokens $\times$ dimensions). In $f = \|z - \bar{z}\|^2 / \|z\|^2$, $\bar{z}$ is the per-dimension token mean within a layer, computed for each of models A and B and averaged. Every metric is computed per layer, and we report the simple average over the 32 layers. The fit R^2 over all layer correspondences in §III is an in-sample ridge linear probe, fit and evaluated on the same 22,400 rows for each layer pair.

b) What f does and does not show.: f is a second moment, not a measure of information, and the inferences it licenses run in one direction only. $f \rightarrow 0$ proves that the latent has collapsed to a constant, and hence that it carries no per-token information at all — detecting this degeneracy is precisely why f was introduced. The converse does not hold, and indeed this appears in the training process of this very study. At iteration 0, after a single update from random initialization, $f=0.926$ — the maximum over the entire training run — yet all four read-outs at that same point are negative (in the order RWKV self-recon/RWKV $\rightarrow$ Pythia/Pythia self-recon/Pythia $\rightarrow$ RWKV, $-0.042/-0.108/-0.029/-0.135$) — that is, worse than predicting the token mean. A large f means only that the per-token latents are spread

out in angular direction; under layer normalization without elementwise gains, f is strictly invariant to scaling. Spread arises equally from meaningful variation and from meaningless variation. Consequently, it is the read-out, not f , that can claim “the varying component carries information”; and since the read-out is a centered R^2 , any token-independent predictor scores exactly 0. $B \rightarrow B = 0.711$ cannot be obtained from any constant. Moreover, the two must be read together. In the latter half of training (85,000 $\rightarrow$ 415,000 iterations), while f increases 0.189 $\rightarrow$ 0.350, ρ_{ctr} falls 0.949 $\rightarrow$ 0.901, because part of the added spread is variation on which the two families do not agree.

c) Raw correlation fails to capture improvement: an example from the training dynamics.: Over iterations 58,000 $\rightarrow$ 66,000, after the learning rate has dropped, f increases 0.138 $\rightarrow$ 0.153, and ρ_{ctr} (0.950 $\rightarrow$ 0.952) and the four read-outs (RWKV self-recon 0.562 $\rightarrow$ 0.570, RWKV $\rightarrow$ Pythia 0.477 $\rightarrow$ 0.486, Pythia self-recon 0.545 $\rightarrow$ 0.552, Pythia $\rightarrow$ RWKV 0.504 $\rightarrow$ 0.514) all improve together. Yet over the same interval ρ_{raw} goes from 0.9931 to 0.9927 — if anything, a slight decrease. Even as the model replaces the common mode with token-dependent information that can be read out, the raw correlation captures none of the improvement. This exemplifies exactly why alignment quality must not be measured with raw correlation.

d) Trajectory of f in the wide latent.: Because nearly all of the latent’s energy can fit into the common mode, the wide latent collapses early in training to $f \approx 0.02$ — 98% of the latent being a token-independent constant — ($f=0.015$ at 1,000 iterations). Recovery from there is slow: even at 25,000 iterations f is only 0.058, and at 85,000 iterations still 0.189; the deepest point of recovery, 0.350, is reached only in the final stage of training. Even at that value, 65% of the latent’s energy is a shared constant, and only the remaining 35% can carry per-token information.